

TN-2020

Grade 12

Physics Part-III

Time: 3 Hours

Max. Marks: 70

Instructions:

- (1) Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.
- (2) Use Blue or Black ink to write and underline and pencil to draw diagrams.

PART-I

15 × 1 = 15

Note: (i) Answer all the questions.

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer

1. The frequency range of 30 MHz to 400 GHz is used for
 - (a) Satellite communication
 - (b) Ground wave propagation
 - (c) Space wave propagation
 - (d) Sky wave propagation
2. In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is:
 - (a) Q

- (b) $\frac{Q}{2}$
 (c) $\frac{Q}{\sqrt{3}}$
 (d) $\frac{Q}{\sqrt{2}}$

3. Type of material which emits white light in LED:

- (a) GaInN
 (b) SiC
 (c) AlGaP
 (d) GaAsP

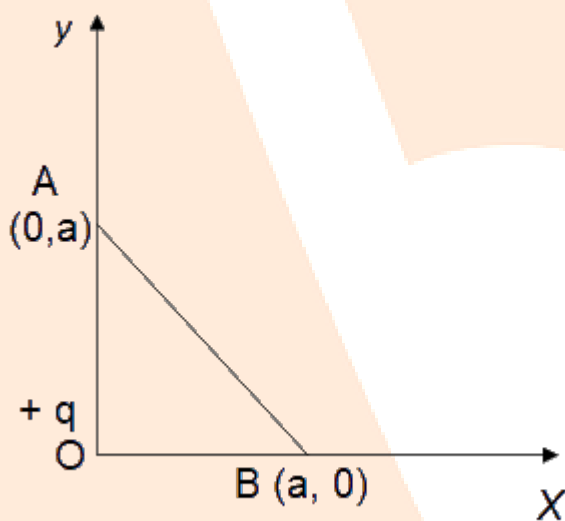
4. A particle of mass m , carrying charge q is accelerated through a potential of V (Volt). When this accelerated charge comes under the influence of perpendicular magnetic field, the force acting on it is :

- (a) $\sqrt{\frac{2q^3BV}{m^3}}$
 (b) $\sqrt{\frac{2q^3BV}{m}}$
 (c) $\sqrt{\frac{q^3B^2V}{2m}}$
 (d) $\sqrt{\frac{2q^3B^2V}{m}}$

5. Charging current for a capacitor is 0.2 A, find the displacement current

- (a) Zero
 (b) 0.2 A
 (c) 0.4 A

- (d) 0.1 A
6. In Bohr Atom Model when the principal quantum number (n) increases the velocity of electron:
- (a) increases and then decreases
 - (b) increases
 - (c) decreases
 - (d) remains constant
7. In the given diagram a point charge + q is placed at the origin O. Work done in taking another point charge - Q from point A to point B is:



- (a) $\frac{qQ}{4\pi\epsilon_0 a^2} \left(\frac{q}{\sqrt{2}} \right)$
- (b) Zero
- (c) $\left[\frac{-qQ}{4\pi\epsilon_0} \frac{1}{a^2} \right] \sqrt{2}a$

(d) $\left[\frac{qQ}{4\pi\epsilon_0} \frac{1}{a^2} \right] \sqrt{2}a$

8. The nucleus is approximately spherical in shape . Then the surface area of nucleus having mass number A varies as:

(a) $A^{\frac{5}{3}}$

(b) $A^{\frac{2}{3}}$

(c) $A^{\frac{4}{3}}$

(d) $A^{\frac{1}{3}}$

9. Two light waves from slit S1 and S2 on reaching points P and Q on a screen in Young's double slit experiment have a path difference zero and $\frac{\lambda}{4}$ respectively. The ratio of light intensities at P and Q will be :

(a) 4:1

(b) 3:2

(c) $\sqrt{2}:1$

(d) 2:1

10. The radius of curvature of curved surface at a thin planoconvex lens is 10 cm and the refractive index is 1.5. If the plane surface is silvered then the focal length will be:

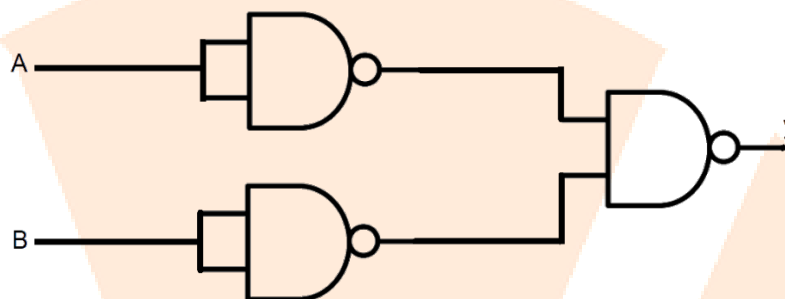
(a) 20 cm

(b) 5 cm

(c) 10 cm

(d) 15 cm

11. The given electrical network is equivalent to:



(a) NAND gate

(b) OR gate

(c) NOT gate

(d) Ex-OR

12. Magnetic field at any point at any distance R due to a long straight conductor carrying current varies as

(a) R^2

(b) R

(c) $\frac{1}{R^2}$

(d) $\frac{1}{R}$

13. If voltage applied on a capacitor is increased from V to $2V$, choose the correct conclusion.

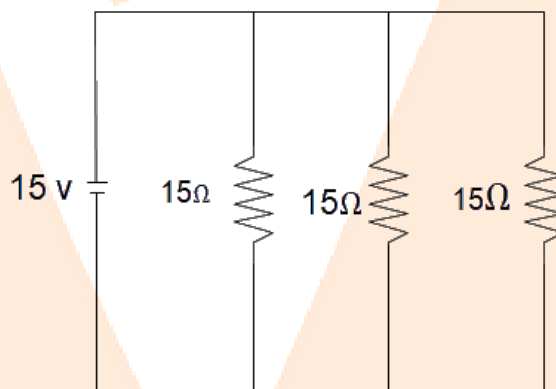
(a) Both Q and C remain the same

(b) Q remains the same, C is doubled

- (c) Q is doubled, C is doubled
(d) C remains the same, Q is doubled
14. A light of wavelength 500 nm is incident on a sensitive plate of photoelectric work function 1235 eV. The kinetic energy of the photo electrons emitted is :

(Take $h = 6.6 \times 10^{-34} \text{ Js}$)

- (a) 1.16 eV
(b) 0.58 eV
(c) 2.48 eV
(d) 1.24 eV
15. The current in the circuit is :



- (a) 1A
(b) 1A
(c) 2A
(d) 3A

PART-II

 $6 \times 2 = 12$

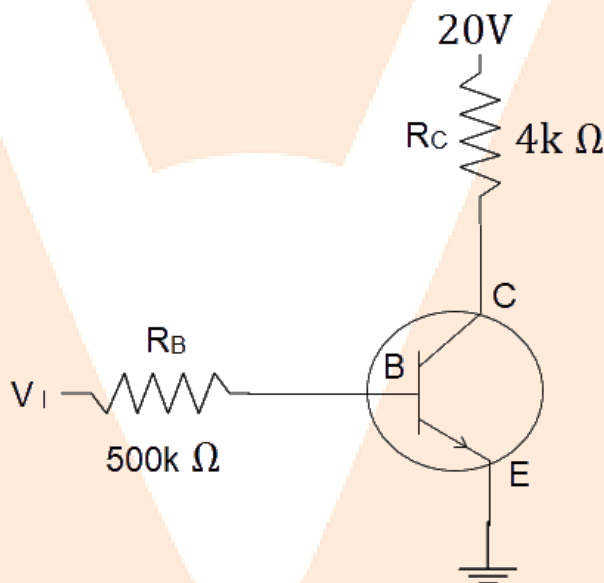
Note: Answer any six question. Question number 24 is compulsory

16. What do you mean by doping?
17. What are the uses of X-rays?
18. An ideal transformer has 460 and 40,000 turns in the primary and secondary coils respectively. Find the voltage developed per turn of the secondary coil if the transformer is connected to a 230 V AC main.
19. Distinguish between Fresnel and Fraunhofer type of diffraction
20. What is corona discharge?
21. What is skip area?
22. What are the properties of neutrino?
23. Two materials X and Y are magnetised whose intensity of magnetisation are 500 Am^{-1} and 2000 Am^{-1} respectively. The magnetizing field is 1000 Am^{-1} . What is the ratio between the susceptibilities of the two material?
24. Why electron is preferred Over X-ray in microscope?

PART -III

6 × 3 = 18

25. Explain the conversion of galvanometer into voltmeter.
26. The resistance of a nichrome wire at 0°C is $10\ \Omega$. If its temperature coefficient of resistance is $0.004/^{\circ}\text{C}$, find its resistance at boiling point of water. Comment on the result.
27. What are the important inferences from the average binding energy curve?
28. In the circuit shown in the figure, the input voltage V_i is 20 V, $V_{BE} = 0\text{ V}$ and $V_{CE} = 0\text{ V}$, What are the values of I_B , I_C and β



29. Derive the expression for equivalent capacitance, when capacitor are connected in parallel
30. What are advantages and disadvantages of AC over DC?

31. Two light sources of equal amplitude interfere with each other, Calculate the ratio of maximum and minimum intensities.
32. Derive an expression for de-Broglie wavelength of electrons.
33. Modulation help to reduce the antenna size in wireless communication – Explain

PART- IV

5 × 5 = 25

Note: Answer all the questions.

34. (a) Obtain the expression for the induced emf by changing relative orientation of the coil with the magnetic field (Graph not necessary).

OR

- (b) Derive the mirror equation and the equation for lateral magnification.

35. (a) Deduce the expression for the force between two long parallel current carrying conductors.

OR

- (b) Write down Maxwell equations in integral form

36. (a) Describe Davisson - Germer experiment Which demonstrated the wave nature of electrons.

OR

- (b) (i) Derive an expression for the orbital energy of an electron in hydrogen atom using Bohr theory.
- (ii) An electron in Bohr's hydrogen atom has an energy of - 3.4 eV. What is the angular momentum of the electron?
37. (a) Explain the Working of the transistor as an oscillator.
- (b) Find out the phase relationship between voltage and current in a pure inductive circuit
38. (a) State Gauss Law in electrostatics. Obtain an expression for Electric field due to an infinitely long charged wire

OR

- (b) How the emf of two cells are compared using potentiometer?